SEFI Canonical Core
Warp-Metric, Operators, SEFI:DNA, and Unified Evolution

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SEFI Canon

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Preface

This document records the canonical, minimal mathematical definitions for SEFI's core structures: the warp-metric formalization, the canonical operator set, a minimal SEFI:DNA geometric model, and the unified manifold evolution equation used for SEFI cosmology. Each definition is presented in a compact, explicit form suitable for implementation, analysis, and further refinement. These forms are chosen to be minimal, physically interpretable, and mathematically well-posed.

1 Notation and basic assumptions

Let $\mathcal{M}$ be a smooth n-dimensional manifold with local coordinates x^μ ($\mu = 0, \ldots, n-1$). Let $g_{\mu\nu}(x)$ denote a smooth reference metric on $\mathcal{M}$ (signature convention left general). Covariant derivatives with respect to $g_{\mu\nu}$ are written ∇_μ. Scalar Laplacian is $\Delta \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu$. Bold symbols denote tensor fields when helpful.

2 Canonical SEFI warp-metric

Definition

Introduce a scalar warp field $W(x) \in C^\infty(\mathcal{M})$ and a small set of coupling constants $\alpha, \beta \in \mathbb{R}$. Define the canonical SEFI warp correction functional

$$F_{\mu\nu}[W] \equiv \alpha\, W\, g_{\mu\nu} + \beta\, \nabla_\mu \nabla_\nu W.$$

The **canonical SEFI warp-metric** is

$$\tilde{g}_{\mu\nu}(x) \equiv g_{\mu\nu}(x) + F_{\mu\nu}[W(x)].$$

Remarks and minimal example

- α controls an isotropic conformal-like scaling by W.
- β introduces local anisotropic curvature via second derivatives of W.
- For a static, spherically symmetric warp $W(r)$ on a background $g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \dots)$, $\tilde{g}$ is obtained by substituting $W(r)$ into $F_{\mu\nu}$ and computing Christoffel symbols and curvature tensors in the usual way.

3 Canonical SEFI operator set

Let $\Phi(x)$ denote a generic SEFI field (scalar, vector, or tensor as context requires). The canonical operators are defined as follows.

Warp operator $\hat{W}$

For scalar fields,

$$\hat{W}\,\Phi \equiv W(x)\,\Phi(x).$$

For tensor fields, $\hat{W}$ acts by pointwise multiplication on each component in a chosen frame.

Tension operator $\hat{T}$

Introduce a canonical tension vector field $T^\mu(x)$. Define

$$\hat{T}\,\Phi \equiv \nabla_\mu\big(T^\mu(x)\,\Phi(x)\big).$$

For vector/tensor Φ the divergence is taken componentwise with index contraction consistent with tensor rank.

Coherence operator $\hat{C}$

Introduce a scalar coherence density $C(x) \geq 0$. Define

$$\hat{C}\,\Phi \equiv C(x)\,\Phi(x).$$

Correction (restoration) operator $\hat{R}$

Define a linear restorative operator combining local relaxation toward an equilibrium configuration $\Phi_{\text{eq}}(x)$ and diffusive smoothing:

$$\hat{R}\,\Phi \equiv -\gamma\big(\Phi - \Phi_{\text{eq}}\big) + \kappa\,\Delta\Phi,$$

with $\gamma > 0$ (relaxation rate) and $\kappa \geq 0$ (diffusion coefficient). Nonlinear generalizations may be introduced later; this linear form is canonical for the base theory.

4 Minimal SEFI:DNA geometric model

Model living systems as submanifolds $\mathcal{B} \subset \mathcal{M}$ carrying a state field $\Psi(x,t)$ (scalar or low-rank tensor) describing structural/functional geometry.

State evolution

The canonical SEFI:DNA evolution equation combines restoration, duplication (growth), and curvature-dependent aging:

$$\partial_t \Psi = \hat{R}\Psi + \hat{D}\Psi - \hat{A}\Psi.$$

Canonical forms

Choose minimal explicit forms:

$$4pt]\hat{D}\Psi = \sigma\,\Psi\left(1 - \frac{\Psi}{\Psi_{\max}}\right), \qquad \sigma \geq 0,\ \Psi_{\max} > 0,$$

$$4pt]\hat{A}\Psi = \mu\,\mathcal{F}\big(\mathcal{K}[\tilde{g}]\big)\,\Psi, \qquad \mu \geq 0.$$

Here $\mathcal{K}[\tilde{g}]$ denotes a scalar curvature measure derived from the warp-metric (for example the Ricci scalar $R[\tilde{g}]$ or a localized curvature density), and $\mathcal{F}$ is a nonnegative function mapping curvature to an aging rate (canonical choice $\mathcal{F}(u) = \max(0, u)$ or a smooth approximation).

Interpretation

- Restoration $\hat{R}$ models healing and local equilibration.
- Duplication $\hat{D}$ is logistic growth limited by $\Psi_{\max}$.
- Aging $\hat{A}$ couples geometric fatigue to curvature: regions of high curvature increase effective aging.

5 Canonical unified manifold evolution (SEFI cosmology)

Treat the global SEFI manifold $\mathcal{U}$ with time-dependent warp-metric $\tilde{g}_{\mu\nu}(x, t)$. SEFI replaces a purely stress–energy driven evolution with a unified evolution functional that includes warp, tension, and coherence source contributions.

General evolution equation

Define the canonical evolution equation

$$\partial_t \tilde{g}_{\mu\nu} = -2\kappa_G\big(R_{\mu\nu}[\tilde{g}] - \tfrac{1}{2}R[\tilde{g}]\,\tilde{g}_{\mu\nu}\big) + S_{\mu\nu}[W, T, C],$$

where $R_{\mu\nu}[\tilde{g}]$ and $R[\tilde{g}]$ are the Ricci tensor and scalar of $\tilde{g}$, $\kappa_G > 0$ is a geometric coupling constant, and $S_{\mu\nu}$ is the canonical SEFI source tensor built from warp, tension, and coherence fields. A minimal canonical choice for $S_{\mu\nu}$ is

$$S_{\mu\nu} \equiv \lambda_W \left(\nabla_\mu \nabla_\nu W - g_{\mu\nu} \Delta W\right) + \lambda_T \left(\nabla_\mu T_\nu + \nabla_\nu T_\mu\right) + \lambda_C \, C \, \tilde{g}_{\mu\nu},$$

with constants $\lambda_W, \lambda_T, \lambda_C \in \mathbb{R}$.

Homogeneous isotropic reduction (canonical FLRW analogue)

Assume spatial homogeneity and isotropy with metric ansatz (cosmological time t)

$$\tilde{g} = -\mathrm{d}t^2 + a(t)^2 \, \gamma_{ij} \, \mathrm{d}x^i \mathrm{d}x^j,$$

where γ_{ij} is a constant curvature spatial metric and $a(t)$ the scale function. Under the canonical source choice and spatially uniform fields $W(t), T^i = 0, C(t)$, the evolution reduces to an ODE for $a(t)$:

$$\frac{\dot{a}}{a} \equiv H(t), \qquad \dot{H} + H^2 = -\frac{\kappa_G}{(n-1)} \big(R_{00}[\tilde{g}]\big) + \frac{1}{2(n-1)} S^0{}_0,$$

which can be written in the canonical SEFI Friedmann-like form

$$H^2 = \frac{1}{(n-1)(n-2)} \Big(\rho_{\mathrm{SEFI}}(W, C) - k \, a^{-2}\Big),$$

where $\rho_{\mathrm{SEFI}}(W, C)$ is an effective geometric energy density derived from $S_{\mu\nu}$ and k is the spatial curvature constant. The explicit mapping from $S_{\mu\nu}$ to ρ_{SEFI} follows standard reduction procedures for homogeneous metrics.

Remarks

- The evolution equation is intentionally written in a form that mirrors geometric flows (Ricci flow style) plus SEFI source terms; this makes analytic and numerical study tractable.
- Constants $\kappa_G, \lambda_W, \lambda_T, \lambda_C$ are canonical parameters to be fixed by phenomenology or chosen as units for model exploration.

6 Canonical examples and minimal parameter choices

For initial canonical studies adopt the following minimal parameter set:

$$\alpha = 1, \quad \beta = \tfrac{1}{2}, \quad \gamma = 1, \quad \kappa = 10^{-2}, \quad \sigma = 0.5, \quad \Psi_{\max} = 1, \quad \mu = 10^{-3},$$

$$\kappa_G = 1, \quad \lambda_W = 1, \quad \lambda_T = 0.5, \quad \lambda_C = 0.1.$$

These values are placeholders chosen to ensure numerical stability in exploratory simulations; they are canonical starting points for v0 analysis.

7 Implementation notes

- Compute geometric quantities (Christoffel symbols, Ricci tensor, scalar) from $\tilde{g}_{\mu\nu}$ using standard symbolic or numeric tensor libraries.
- For numerical evolution use implicit or semi-implicit time integrators to handle stiffness introduced by second derivatives in $F_{\mu\nu}$ and diffusion terms in $\hat{R}$.
- Boundary conditions: for compact manifolds use periodic or fixed boundary conditions; for open domains use absorbing or radiative boundary conditions consistent with the physical scenario.
- Validate SEFI:DNA dynamics by testing conservation/monotonicity properties (e.g., total mass or integrated coherence) under controlled initial conditions.

Conclusion

The canonical forms above provide a compact, implementable mathematical backbone for SEFI canon. They are intentionally minimal, physically interpretable, and numerically tractable. Subsequent canonical refinements should preserve the structural roles of warp, tension, coherence, and correction while improving functional forms and coupling choices based on analytic insight and empirical constraints.